

THE STORY OF GEOMETRY

ONE OF THE OLDEST BRANCHES OF MATHEMATICS, GEOMETRY IS EXPANDING INTO NEW AREAS

The term “geometry” derives from ancient Greek words meaning “Earth measurement,” but this branch of mathematics encompasses more than map-making. It is about relationships between size, shape, and dimension—and also about the nature of numbers and mathematics itself.

Geometry first arose as a series of ad hoc rules and formulas used in planning, construction, and mathematical problem-solving across the ancient world. Greek philosophers such as Thales, Pythagoras, and Plato were the first to recognize geometry’s fundamental relationship to the nature of space, and to establish it as a field of mathematics worthy of study in its own right. Euclid, probably a student of Plato and a teacher at Alexandria, summed up early Greek geometry in his great work *The Elements*, written around 300 BCE, and established fundamental mathematical and

scientific principles through complex geometrical models developed from a handful of simple rules or axioms.

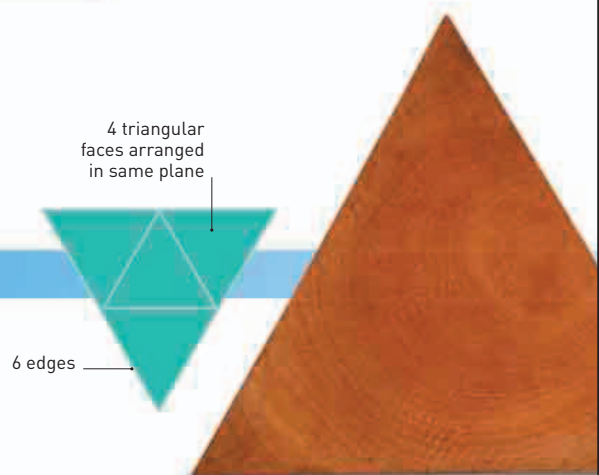


Axioms of geometry
Euclid's approach to geometry had a huge and lasting influence on later mathematicians.

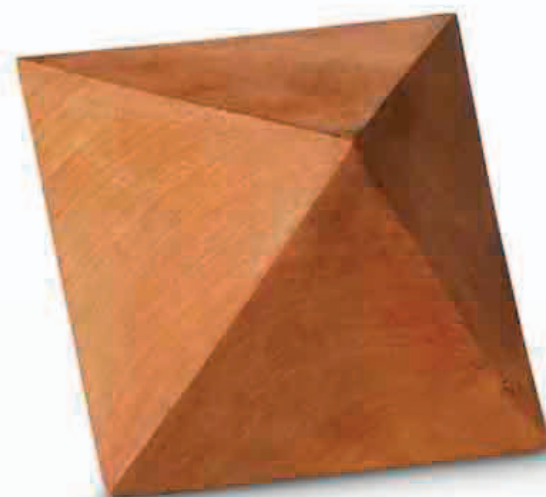
BREAKTHROUGHS IN UNDERSTANDING

Throughout medieval times, philosophers and mathematicians from various cultures continued to use geometry in their models of the Universe, but the next major breakthrough did not come until the 17th century, with the work of French mathematician and philosopher René Descartes. His invention of coordinate systems to describe the positions of points in two-dimensional and three-dimensional space gave rise to the field of analytical geometry, which used the new tools of mathematical algebra to describe and solve geometrical problems.

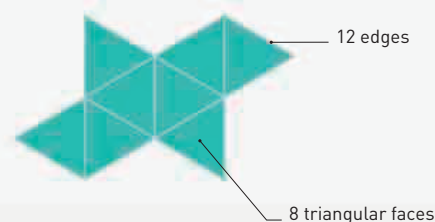
Descartes's work led to more exotic forms of geometry. Mathematicians had long known that there were regions, such as the surface of a sphere, where the axioms of Euclidian geometry did not hold. Investigation of such non-Euclidian geometries revealed even more fundamental principles linking geometry and number, and in 1899 allowed German mathematician David Hilbert to produce a new, more generalized, set of axioms. Throughout the 20th century, and into the 21st, these have been applied to a huge variety of mathematical scenarios.



Tetrahedron



Octahedron



c.2500 BCE

Practical geometry
Early geometry is driven by the need to solve problems such as working out the volume of material required to build a pyramid.



Pyramids at Giza

360 BCE

Platonic solids

These five regular, convex polyhedra (solids with several sides) are long known, but Plato now links them to ideas about the structure of matter. They comprise five shapes that can be formed by the joining together of identical faces along their edges.

c.400 CE

“Archimedean” solids
Greek mathematician Pappus describes 13 convex polyhedra, comprising regular polygons of two or more types meeting in identical vertices or corners.

1619
Kepler's polyhedra
German mathematician Johannes Kepler discovers a new class of polyhedra known as star polyhedra.



c.500 BCE

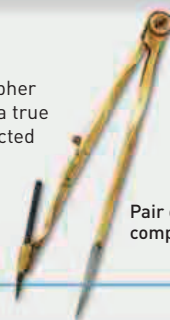
Pythagoras
The Greek philosopher lends his name to the formula for calculating the hypotenuse (long side) of a right-angled triangle from the lengths of its other two sides.



Theorem of Pythagoras

4th century BCE

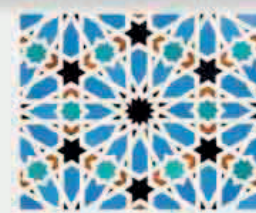
Geometric tools
The hugely influential philosopher Plato argues that the tools of a true geometrician should be restricted to the compass and straight edge, and so helps establish geometry as a science rather than a practical craft.



Pair of compasses

9th century

Islamic geometry
Mathematicians and astronomers of the Islamic world explore the possibilities of spherical geometry; geometric patterns used in Islamic decoration at this time show similarities to modern fractal geometry.



Mosaic at Alhambra